C PROGRAMMING – ASSIGNMENT 3

Roll No: 2025115054

1) Write a C program to store n elements in an array using dynamic memory allocation and find the largest number.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
 int n, i, largest;
 int *arr;
 printf("Enter n: ");
 scanf("%d", &n);
 arr = (int *)malloc(n * sizeof(int));
 printf("Enter %d elements: ", n);
 for (i = 0; i < n; i++)
 {
 scanf("%d", arr + i);
 }
 largest = *arr;
 for (i = 1; i < n; i++)
 {
 if (*(arr + i) > largest)
 {
 largest = *(arr + i);
 }
 }
 printf("\nLargest number: %d", largest);
 free(arr);
 return 0;
}
```

Output:

```
Enter n: 6
Enter 6 elements: 18 72 35 91 44 63
Largest number: 91
```

2) Print an array in reverse order using pointers.

Source Code:

```
#include <stdio.h>
int main()
{
 int arr[10], n, i;
 int *p;
 printf("Enter n: ");
 scanf("%d", &n);
 p = arr;
 printf("Enter %d elements: ", n);
 for (i = 0; i < n; i++)
 {
 scanf("%d", p + i);
 }
 printf("\nArray in reverse order: ");
 for (i = n - 1; i >= 0; i--)
 {
 printf("%d ", *(p + i));
 }
 return 0;
}
```

Output:

```
Enter n: 6
Enter 6 elements: 11 22 33 44 55 66
Array in reverse order: 66 55 44 33 22 11
```

3) Write a C program to store information of students using structure. Sort items by name.

Source Code:

```
#include <stdio.h>
#include <string.h>
struct student
{
 int rollno;
 char name[20];
 float mark;
};
int main()
{
 struct student s[10], temp;
 int n, i, j;
 printf("Enter n: ");
 scanf("%d", &n);
 printf("\n--- Student Details ---\n");
 for (i = 0; i < n; i++)
 {
 printf("\nRoll no: ");
 scanf("%d", &s[i].rollno);
 printf("Name: ");
 scanf("%s", s[i].name);
 printf("Mark: ");
 scanf("%f", &s[i].mark);
 }
 for (i = 0; i < n - 1; i++)
 {
 for (j = i + 1; j < n; j++)
 {
 if (strcmp(s[i].name, s[j].name) > 0)
 {
 temp = s[i];
 s[i] = s[j];
 s[j] = temp;
 }
 }
 }
 printf("\n--- Sorted by Name ---\n");
 for (i = 0; i < n; i++)
 {
 printf("\nRoll no: %d", s[i].rollno);
 printf("\nName: %s", s[i].name);
 printf("\nMark: %.2f\n", s[i].mark);
 }
 return 0;
}
```

Output:

```
Enter n: 3
--- Student Details ---
Roll no: 111
Name: Ravi
Mark: 84
Roll no: 112
Name: Arjun
Mark: 92
Roll no: 113
Name: Meena
Mark: 79
--- Sorted by Name ---
Roll no: 112
Name: Arjun
Mark: 92.00
Roll no: 113
Name: Meena
Mark: 79.00
```

Roll no: 111
Name: Ravi
Mark: 84.00

4) Write a C program to add n distances given in inch-feet system using structures.

Source Code:

```
#include <stdio.h>
struct distance
{
 int feet;
 float inch;
};
int main()
{
 struct distance d[10], sum = {0, 0};
 int n, i;
 printf("Enter n: ");
 scanf("%d", &n);
 printf("\n--- Enter Distances ---\n");
 for (i = 0; i < n; i++)
 {
 printf("\nDistance %d", i + 1);
 printf("\nFeet: ");
 scanf("%d", &d[i].feet);
 printf("Inch: ");
 scanf("%f", &d[i].inch);
 sum.feet += d[i].feet;
 sum.inch += d[i].inch;
 }
 if (sum.inch >= 12)
 {
 sum.feet += (int)(sum.inch / 12);
 sum.inch = sum.inch - ((int)(sum.inch / 12) * 12);
 }
 printf("\nTotal Distance: %d feet %.2f inches", sum.feet, sum.inch);
 return 0;
}
```

Output:

```
Enter n: 3
--- Enter Distances ---
Distance 1
Feet: 4
Inch: 9
Distance 2
Feet: 3
Inch: 7
Distance 3
Feet: 2
Inch: 6
Total Distance: 11 feet 10.00 inches
```

5) Write a C program to add two complex numbers by passing structures to a function.

Source Code:

```
#include <stdio.h>
struct complex
{
 float real;
 float imag;
};
struct complex add(struct complex c1, struct complex c2)
{
 struct complex result;
 result.real = c1.real + c2.real;
 result.imag = c1.imag + c2.imag;
 return result;
}
int main()
{
 struct complex c1, c2, result;
 printf("Enter real and imaginary parts of first complex number: ");
 scanf("%f %f", &c1.real, &c1.imag);
 printf("Enter real and imaginary parts of second complex number: ");
 scanf("%f %f", &c2.real, &c2.imag);
 result = add(c1, c2);
 printf("\nSum = %.2f + %.2fi", result.real, result.imag);
 return 0;
}
```

Output:

```
Enter real and imaginary parts of first complex number: 7 3
Enter real and imaginary parts of second complex number: 5 6
Sum = 12.00 + 9.00i
```

6) Write a C program to maintain inventory management system using structures.

Source Code:

```
#include <stdio.h>
struct item
{
 int id;
 char name[20];
 int quantity;
 float price;
};
int main()
{
 struct item item[10];
 int n, i, choice, id, found;
 printf("Enter number of items: ");
 scanf("%d", &n);
 for (i = 0; i < n; i++)
 {
 printf("\nItem %d", i + 1);
 printf("\nItem ID: ");
 scanf("%d", &item[i].id);
 printf("Item Name: ");
 scanf("%s", item[i].name);
 printf("Quantity: ");
 scanf("%d", &item[i].quantity);
 printf("Price: ");
 scanf("%f", &item[i].price);
 }
 do
 {
 printf("\n--- Inventory Management ---\n");
 printf("1. Display Items\n");
 printf("2. Search Item\n");
 printf("3. Update Quantity\n");
 printf("4. Exit\n");
 printf("Enter choice: ");
 scanf("%d", &choice);
 if (choice == 1)
 {
 printf("\n--- Inventory Details ---\n");
 for (i = 0; i < n; i++)
 {
 printf("\nItem ID: %d", item[i].id);
 printf("\nName: %s", item[i].name);
 printf("\nQuantity: %d", item[i].quantity);
 printf("\nPrice: %.2f\n", item[i].price);
 }
 }
 else if (choice == 2)
 {
 printf("\nEnter Item ID: ");
 scanf("%d", &id);
 found = 0;
 for (i = 0; i < n; i++)
 {
 if (item[i].id == id)
 {
 printf("\nItem Name: %s", item[i].name);
 printf("\nQuantity: %d", item[i].quantity);
 printf("\nPrice: %.2f\n", item[i].price);
 found = 1;
 }
 }
 if (found == 0)
 {
 printf("\nItem not found.");
 }
 }
 else if (choice == 3)
 {
```

```
 printf("\nEnter Item ID: ");
 scanf("%d", &id);
 found = 0;
 for (i = 0; i < n; i++)
 {
 if (item[i].id == id)
 {
 printf("Enter new quantity: ");
 scanf("%d", &item[i].quantity);
 printf("\nQuantity updated successfully.");
 found = 1;
 }
 }
 if (found == 0)
 {
 printf("\nItem not found.");
 }
 }
 } while (choice != 4);
 return 0;
}
```

Output:

```
Enter number of items: 2
Item 1
Item ID: 201
Item Name: Pencil
Quantity: 35
Price: 8
Item 2
Item ID: 202
Item Name: Notebook
Quantity: 18
Price: 45
--- Inventory Management ---
1. Display Items
2. Search Item
3. Update Quantity
4. Exit
Enter choice: 1
--- Inventory Details ---
Item ID: 201
Name: Pencil
Quantity: 35
Price: 8.00
Item ID: 202
Name: Notebook
Quantity: 18
Price: 45.00
--- Inventory Management ---
1. Display Items
2. Search Item
3. Update Quantity
4. Exit
Enter choice: 4
```

7) Write a C program to convert a decimal number to binary, octal and hexadecimal.

Source Code:

```
#include <stdio.h>
int main()
{
 int n;
 printf("Enter a decimal number: ");
 scanf("%d", &n);
 printf("\nBinary: ");
 if (n == 0)
 {
 printf("0");
 }
 else
 {
 int binary[32], i = 0, temp = n;
 while (temp > 0)
 {
 binary[i] = temp % 2;
 temp = temp / 2;
 i++;
 }
 while (i > 0)
 {
 i--;
 printf("%d", binary[i]);
 }
 }
 printf("\nOctal: %o", n);
 printf("\nHexadecimal: %X", n);
 return 0;
}
```

Output:

```
Enter a decimal number: 36
Binary: 100100
Octal: 44
Hexadecimal: 24
```

8) Write a C program to calculate difference between two time periods using structures.

Source Code:

```
#include <stdio.h>
struct time
{
 int hour;
 int minute;
 int second;
};
int main()
{
 struct time t1, t2, diff;
 printf("Enter first time (hour minute second): ");
 scanf("%d %d %d", &t1.hour, &t1.minute, &t1.second);
 printf("Enter second time (hour minute second): ");
 scanf("%d %d %d", &t2.hour, &t2.minute, &t2.second);
 if (t1.second < t2.second)
 {
 t1.second += 60;
 t1.minute--;
 }
 diff.second = t1.second - t2.second;
 if (t1.minute < t2.minute)
 {
 t1.minute += 60;
 t1.hour--;
 }
 diff.minute = t1.minute - t2.minute;
 diff.hour = t1.hour - t2.hour;
 printf("\nTime Difference: %d:%d:%d", diff.hour, diff.minute, diff.second);
 return 0;
}
```

Output:

```
Enter first time (hour minute second): 15 40 30
Enter second time (hour minute second): 12 25 15
Time Difference: 3:15:15
```

9) Write a C program to implement employee management system using structures.

Source Code:

```
#include <stdio.h>
struct employee
{
 int id;
 char name[20];
 float salary;
};
int main()
{
 struct employee e[10];
 int n, i, choice, id, found;
 printf("Enter number of employees: ");
 scanf("%d", &n);
 for (i = 0; i < n; i++)
 {
 printf("\nEmployee %d", i + 1);
 printf("\nEmployee ID: ");
 scanf("%d", &e[i].id);
 printf("Name: ");
 scanf("%s", e[i].name);
 printf("Salary: ");
 scanf("%f", &e[i].salary);
 }
 do
 {
 printf("\n--- Employee Management ---\n");
 printf("1. Display Employees\n");
 printf("2. Search Employee\n");
 printf("3. Exit\n");
 printf("Enter choice: ");
 scanf("%d", &choice);
 if (choice == 1)
 {
 printf("\n--- Employee Details ---\n");
 for (i = 0; i < n; i++)
 {
 printf("\nEmployee ID: %d", e[i].id);
 printf("\nName: %s", e[i].name);
 printf("\nSalary: %.2f\n", e[i].salary);
 }
 }
 else if (choice == 2)
 {
 printf("\nEnter Employee ID: ");
 scanf("%d", &id);
 found = 0;
 for (i = 0; i < n; i++)
 {
 if (e[i].id == id)
 {
 printf("\nEmployee ID: %d", e[i].id);
 printf("\nName: %s", e[i].name);
 printf("\nSalary: %.2f\n", e[i].salary);
 found = 1;
 }
 }
 if (found == 0)
 {
 printf("\nEmployee not found.");
 }
 }
 } while (choice != 3);
 return 0;
}
```

Output:

```
Enter number of employees: 2
```

```
Employee 1
Employee ID: 201
Name: Karthik
Salary: 27000
Employee 2
Employee ID: 202
Name: Divya
Salary: 32500
--- Employee Management ---
1. Display Employees
2. Search Employee
3. Exit
Enter choice: 1
--- Employee Details ---
Employee ID: 201
Name: Karthik
Salary: 27000.00
Employee ID: 202
Name: Divya
Salary: 32500.00
--- Employee Management ---
1. Display Employees
2. Search Employee
3. Exit
Enter choice: 3
```

10) Write a C program to calculate the area of rectangle and triangle using function pointers.

Source Code:

```
#include <stdio.h>
float rectangle(float length, float breadth)
{
 return length * breadth;
}
float triangle(float base, float height)
{
 return 0.5 * base * height;
}
int main()
{
 float length, breadth, base, height;
 float (*area)(float, float);
 printf("Enter length and breadth of rectangle: ");
 scanf("%f %f", &length, &breadth);
 area = rectangle;
 printf("\nArea of rectangle: %.2f", area(length, breadth));
 printf("\n\nEnter base and height of triangle: ");
 scanf("%f %f", &base, &height);
 area = triangle;
 printf("\nArea of triangle: %.2f", area(base, height));
 return 0;
}
```

Output:

```
Enter length and breadth of rectangle: 12 7
Area of rectangle: 84.00
Enter base and height of triangle: 10 9
Area of triangle: 45.00
```